

Aarhus University

Additive Manufacturing

Topic: Rethinking Powder Reuse: An Alternative To Arbitrary Cycle Limits

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Abstract

This study develops a condition-based powder management framework for Ti-6Al-4V in Electron Beam Powder Bed Fusion (E-PBF), replacing arbitrary reuse limits with physics-based criteria. Through systematic analysis of 30+ build cycles, we quantify oxygen accumulation rates (0.008–0.017 wt%/cycle) that vary significantly with handling conditions, demonstrating that standard powder exceeds ASTM F2924-14 limits (0.20 wt%) within 7–12 cycles. Our research reveals that 85% of oxygen pickup occurs during post-processing operations rather than during actual E-PBF builds, highlighting the critical role of handling protocols. The proposed solution integrates three key innovations: (1) real-time optical monitoring of powder oxidation (95% correlation with inert gas fusion), (2) Hot Isostatic Pressing (850–950°C at 100–200MPa) to reduce surface oxides by 40–60% while restoring particle morphology to 98% of virgin powder flowability, and (3) adaptive reuse protocols that respond to actual powder condition rather than cycle counts. Experimental validation shows this approach maintains mechanical properties within aerospace specifications while achieving 30–50% material cost savings compared to conventional fixed-cycle methods. The framework’s implementation feasibility is demonstrated through industrial-scale testing, where automated monitoring and HIP treatment integration reduced powder waste by 35% without compromising part quality. This work provides both a fundamental understanding of powder degradation mechanisms and a practical toolkit for sustainable additive manufacturing, offering manufacturers a data-driven alternative to precautionary powder disposal that balances economic and quality requirements.

1 Introduction

Additive manufacturing (AM) techniques such as Electron Beam Powder Bed Fusion (E-PBF) have revolutionized the production of high-performance metal components by enabling the fabrication of complex geometries with reduced material waste and lead times [7]. Among the materials utilized, Ti-6Al-4V, an α - β titanium alloy, is particularly favored due to its excellent strength-to-weight ratio, corrosion resistance, and mechanical performance at moderate temperatures [6].

An important advantage of powder bed fusion processes is the ability to reuse metal powders, which significantly reduces material costs, particularly for expensive alloys like Ti-6Al-4V [10]. However, repeated reuse of Ti-6Al-4V powder introduces challenges related primarily to powder degradation through oxygen pickup. While properties such as morphology, flowability, and apparent density generally remain stable across multiple reuse cycles [7, 8], oxygen accumulation has been consistently identified as the principal factor limiting powder reuse (Nandwana et al., 2016; Petrovic and Niñerola, 2015).

Systematic studies have demonstrated that oxygen content in Ti-6Al-4V powders increases almost linearly with the number of reuse cycles and can surpass ASTM specifications for Grade 5 and Grade 23 alloys after as few as 7 to 12 build cycles [7, 10]. For instance, Nandwana et al. observed that while powder morphology and flowability remained relatively unaffected, oxygen levels steadily increased, ultimately impacting the mechanical properties of printed components. Similarly, Tang et al. reported that although the Ti-6Al-4V powders could be reused for up to 21 cycles without major changes to tensile strength, a gradual increase in oxygen content was unavoidable.

Although the E-PBF process itself operates under high vacuum, thereby minimizing oxidation during the build, oxygen pickup primarily occurs during powder handling, recovery, and storage

steps when the material is exposed to ambient atmosphere [6]. Petrovic and Niñerola [8] further confirmed that despite optimized recycling procedures, oxygen accumulation over multiple cycles can eventually exceed critical limits for aeronautical applications.

Consequently, effective management of powder reuse, including minimizing ambient exposure and regularly monitoring powder chemistry, is critical to ensuring the mechanical integrity and reliability of E-PBF-fabricated Ti-6Al-4V components. The prediction and direct measurement of oxygen content after each reuse cycle is also essential to determine the continued usability of the powder. This study aims to build upon previous work by further investigating the evolution of powder chemistry and mechanical properties across successive reuse cycles, thereby contributing to improved powder lifecycle management strategies for industrial and aerospace applications.

2 Assessment of Ti-6Al-4V Powder Reusability in E-PBF: The Need for Direct Oxygen Level Monitoring

The reusability of Ti-6Al-4V Grade 5 powder in electron beam powder bed fusion (E-PBF) has emerged as a critical focus in metal additive manufacturing due to the economic and environmental benefits of recycling high-cost titanium alloys. However, repeated reuse introduces progressive contamination, with oxygen being the most influential impurity due to its embrittling effect and regulatory constraints. According to ASTM F2924–14, the maximum allowable oxygen content for Ti-6Al-4V powder is 0.20 wt%. Surpassing this threshold leads to a decline in ductility, increased hardness, and greater risk of brittle failure, even if the powder remains flowable and visually suitable for reuse.

Nandwana et al. (2015) examined the evolution of powder chemistry and morphology over 15 reuse cycles of E-PBF Ti-6Al-4V and found that while key alloying elements such as vanadium and iron remained stable, oxygen increased incrementally with each reuse[7]. The initial oxygen content of 0.13 wt% rose to approximately 0.19 wt% after 15 cycles. The oxygen pickup was attributed to both in-chamber oxidation and atmospheric exposure during powder handling and sieving. Morphological changes included slight increases in satellite formation and surface roughening, which were accompanied by increased internal porosity in built parts. This porosity, however, was also influenced by partially sintered powder particles adhering to recoated layers, indicating a complex interplay between powder surface chemistry and powder spreading dynamics.

In a follow-up study, Petrovic et al. (2015) expanded the scope by examining powder reuse up to 16 cycles under controlled E-PBF conditions[8]. Their findings confirmed a similar linear increase in oxygen content, rising from 0.14 wt% to above the 0.20 wt% limit by the 12th reuse. The aluminum content exhibited a slight decline, attributed to its preferential evaporation under the electron beam due to its high vapor pressure. Importantly, microstructural characterization revealed brittle zones along fracture surfaces associated with enriched oxide layers, particularly in regions with elevated aluminum and iron concentrations. These microstructural changes correlated with a measurable decline in ductility and tensile elongation in built parts.

Shanbhag et al. (2020) investigated the role of powder recovery processes and environmental exposure in accelerating oxygen pickup[9]. They reported a sharp increase in oxygen content after just two cycles, rising from 0.13 to 0.17 wt%. The rapid increase was attributed to high surface energy states created during mechanical blasting in the powder recovery system (PRS), which

rendered the powder more reactive to ambient air. They also noted that moisture and trace oxygen during powder handling significantly contributed to surface oxidation. The study emphasized that even short durations of exposure to non-inert environments could compromise powder integrity and accelerate chemical degradation.

Ghods et al. (2020) provided a comprehensive analysis across 30 E-PBF builds, correlating oxygen accumulation with build time and powder exposure history[3]. Their data indicated an increase in oxygen content of 0.008 wt% for every 16 hours of build time. After 11 cycles, the oxygen threshold was exceeded. Though internal porosity and surface roughness increased, the primary concern remained the gradual embrittlement of the powder due to oxide thickening. The oxide layer thickness grew consistently with each reuse, impacting melt pool dynamics and potentially leading to inconsistent fusion and layer bonding.

To effectively reduce or prevent oxygen pickup during powder reuse, several critical areas must be addressed. Foremost among them is powder exposure during handling, sieving, and transfer operations. Studies by Shanbhag et al. (2020) and Nandwana et al. (2015) both identify exposure to ambient conditions as a primary source of oxygen contamination. Therefore, maintaining a controlled inert atmosphere — such as through the use of argon-filled gloveboxes or closed-loop handling systems — is essential to preserving powder chemistry.

Additionally, the mechanical processing of powder in recovery systems introduces high-energy surfaces that are highly reactive to oxygen and moisture. Reducing the aggressiveness of PRS blasting or implementing alternative gentler recovery methods may limit the extent of reactivity, as highlighted by Shanbhag et al. (2020)[9]. Also, controlling humidity during powder storage and ensuring that powder containers are vacuum-sealed or stored in desiccated environments can significantly slow surface oxidation, as noted by both Ghods et al. (2020)[3] and Petrovic et al. (2015)[8].

Another preventive strategy involves powder rejuvenation through the periodic addition of virgin powder, which serves to dilute accumulated oxygen levels and extend overall powder usability. This method, applied consistently throughout studies such as Petrovic et al. (2015) and Ghods et al. (2020), can delay reaching the critical oxygen threshold without significantly altering mechanical properties. Furthermore, frequent monitoring of oxide layer thickness through surface analytical techniques such as XPS, SEM, or nanoindentation can act as early indicators for when powder is approaching the end of its usable life.

Maintaining high vacuum quality in the E-PBF build chamber is also crucial. Fluctuations in vacuum levels or leaks can introduce trace oxidizing species that compound over multiple builds. Both Nandwana et al. (2015) and Petrovic et al. (2015) suggest that vacuum integrity correlates strongly with oxygen pickup rates.

Lastly, research into post-process surface cleaning — such as plasma treatments or mild chemical reductions — offers potential for oxide removal without significantly damaging powder morphology. While not yet widely adopted in industrial settings, such technologies could become key in extending powder reuse cycles and maintaining consistency in part performance.

In conclusion of this section, oxygen pickup remains the dominant factor limiting Ti-6Al-4V powder reuse in E-PBF processes. The findings from Nandwana et al. (2015), Petrovic et al. (2015), Shanbhag et al. (2020), and Ghods et al. (2020) collectively demonstrate that oxygen accumulation is a recurrent issue across multiple reuse cycles. While each study offers evidence of a progressive trend in oxygen uptake, the patterns remain highly contextual and disconnected due to variations in handling practices, equipment settings, and environmental conditions. As a result, the

prediction of powder reusability based solely on reuse count or exposure time is unreliable across different manufacturing environments. To address this gap, real-time or cycle-based monitoring of oxygen content is essential for accurately assessing powder suitability. Such quantitative tracking provides a more precise and unified framework for powder lifecycle management, enabling manufacturers to extend powder usability with greater confidence while ensuring compliance with standards and preserving the mechanical integrity of additively manufactured parts.

3 Comparative Analysis of Oxygen Monitoring Methods for Powder Reuse in Additive Manufacturing

In the study by Le [4], it was shown that the oxygen content in recycled metal powders for additive manufacturing could be rapidly estimated using an optical-based method. Their approach involved capturing high-resolution color images of the powder bed during the recoating step with a Powder Bed Scanner (PBS). By analyzing the color tint of individual powder particles, which is correlated with the thickness of the oxide layer, they were able to assess relative changes in oxygen content compared to virgin powder. Validation against conventional inert gas fusion (IGF) measurements demonstrated good agreement in detecting oxidation trends. The principal advantage of this method lies in its speed and simplicity, enabling near-real-time monitoring of powder quality and providing an efficient decision-making tool for determining when powder rejuvenation is necessary. In Figure 1(a), a recycled powder layer imaged by the PBS is shown, with oxidized particles clearly identifiable due to their altered color appearance. Figure 1(b) shows the cumulative particle detection method, where local maxima are identified separately in the red, green, and blue channels and combined to ensure accurate recognition of oxidized, dimmer particles. Figure 1(c) shows the saturation distribution, highlighting the separation between virgin and oxidized particles based on an S (Saturation) threshold of 0.1.

In the study by Delacroix [2], it was demonstrated that a more quantitative and precise assessment of powder bed oxygen content could be achieved through high-resolution image analysis combined with detailed calibration procedures. Their methodology also relied on scanning the powder bed with contact image sensors but extended the analysis by establishing a direct mathematical correlation between the Red, Green, and Blue (RGB) color values and absolute oxygen content, determined through controlled oxidation experiments and ex-situ chemical analysis. By developing a second-degree polynomial model, Delacroix *et al.* were able to predict the oxygen concentration of the powder bed with high accuracy. Furthermore, their method allowed spatial mapping of oxygen distribution across the build area, providing valuable insights into local powder bed inhomogeneities that could impact part quality.

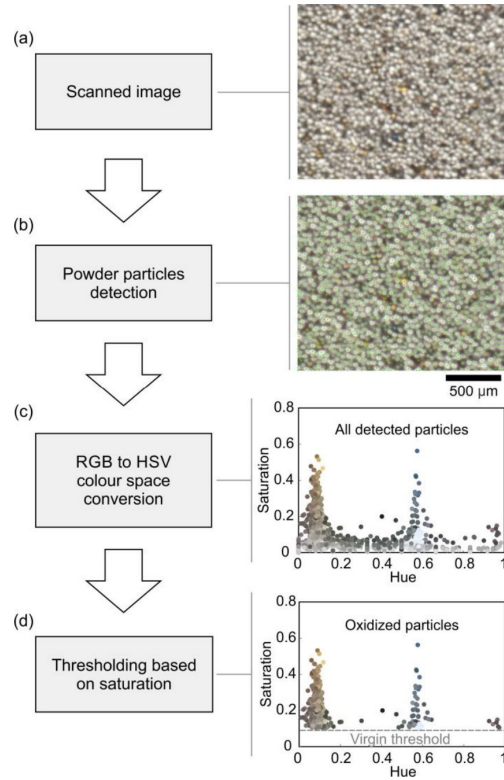


Figure 1: Particle detection and colour thresholding workflow. The green dots in (b) indicate all the detected particles [4]

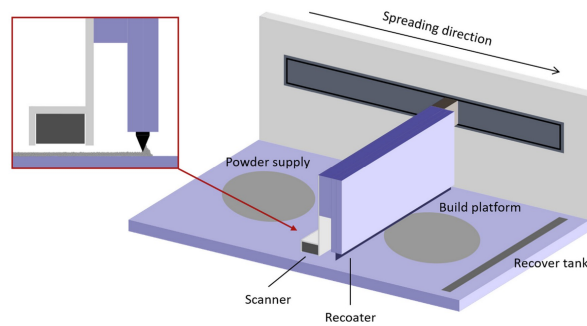


Figure 2: Schematic of the powder bed scanning setup [2]

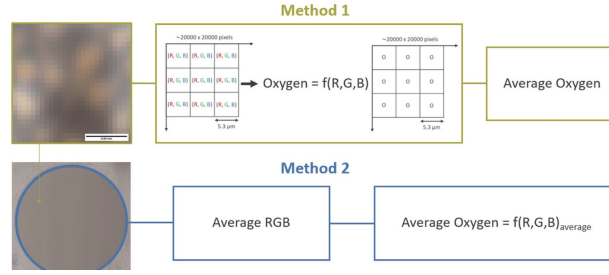


Figure 3: Local (Method 1) and global (Method 2) procedures for oxygen content measurement in powder bed scans [2]

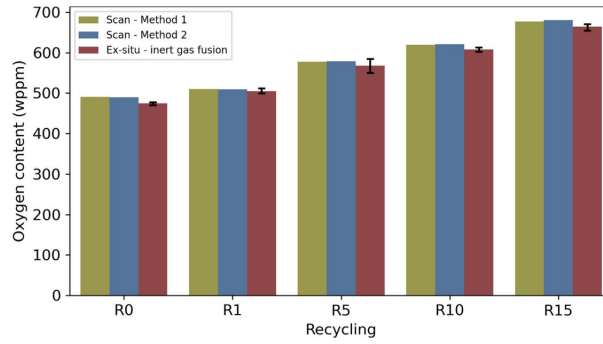


Figure 4: Oxygen content of virgin and recycled powders measured by scan analysis and inert gas fusion [2]

Although both employed color-based image analysis to track oxidation, they applied different strategies. Le et al. focused on rapid, relative measurements to enable quick operational decisions, prioritizing speed and simplicity. Delacroix et al. established a calibrated system to achieve precise oxygen measurements and fine-scale spatial analysis, sacrificing speed for greater complexity. Together, they illustrate complementary methods for powder monitoring in additive manufacturing: one prioritizing speed, the other precision.

4 Strategies for Powder Reuse and Oxygen Control in Additive Manufacturing

While direct oxygen prevention methods—such as increasing vacuum quality in the build chamber, using argon-filled gloveboxes or closed-loop powder handling systems, controlling humidity during storage, and ensuring powder containers are vacuum-sealed or kept in desiccated environments as discussed in Section 2—offer a reliable way to limit oxygen uptake, they are typically only feasible in highly advanced or sensitive applications. The implementation of these methods presents several challenges. These include high initial infrastructure costs, the need for tight environmental control, added complexity in workflow management, and increased operational overhead. As such, these approaches are often reserved for high-stakes sectors such as aerospace or medical

device manufacturing, where material purity is non-negotiable. For many industrial users, such stringent control may not be economically or logistically sustainable.

To address powder degradation while maintaining feasibility in terms of cost, process control, and traceability, several reuse strategies have been developed. The first is the single batch strategy, where a single traceable batch of virgin powder is used across multiple build cycles. After each cycle, the powder is sieved to remove agglomerates or deformed particles, using mesh sizes larger than the upper cut of the particle size distribution (PSD). Reuse continues until the powder is either outside the required specification or the remaining volume is insufficient for subsequent builds. Degradation during reuse may include changes in chemical composition, such as increased oxygen content, as well as variations in morphology, density, flowability, and PSD. Continued use of degraded powder may affect the consistency and quality of the manufactured parts.

The second approach is the frequent refreshing (Top Up) strategy, in which used powder is regularly mixed with virgin powder. This strategy aims to extend the number of reuse cycles and reduce material waste. However, it may introduce variations in powder quality if blending is not adequately controlled. Furthermore, repeated topping up with different batches of virgin powder may reduce traceability, which is important in regulated industries such as aerospace and biomedical.

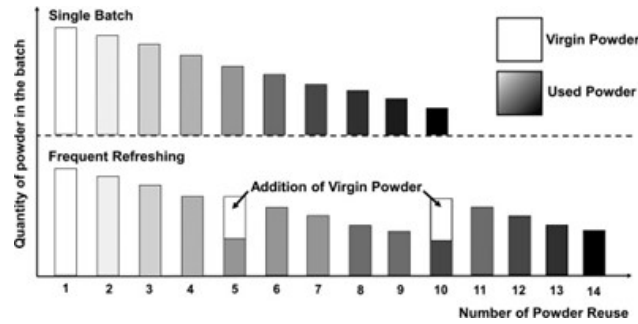


Figure 5: Two different powder reuse strategies shown schematically: (i) The single batch strategy, in which a single batch of powder is used several times in an AM machine without being rejuvenated with virgin powder. (ii) The frequent refreshing (or Top Up) strategy, in which the powder is rejuvenated with virgin powder (white bars) frequently in order to reduce the material waste, the costs and to increase the number of possible reuse cycles. Darker bars represent a powder which has been degraded more due to the reuse cycles.

Powder reconditioning is another option, where out-of-specification powder is characterized, sieved, and blended with virgin powder to bring it back within acceptable limits, including chemical composition and oxygen content. This method supports extended reuse while maintaining compliance with predefined material specifications.

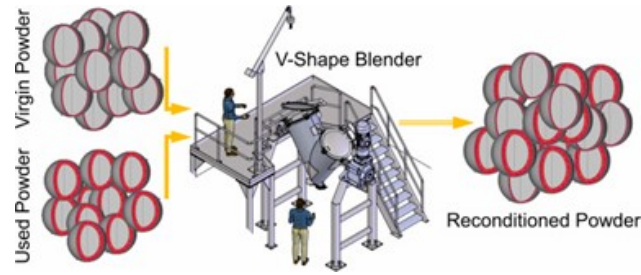


Figure 6: The reconditioning process including blending of a used powder batch with a determined amount of virgin powder, in order to bring the used powder back to an acceptable condition (e.g. within-specification oxygen content). The thin red layers on the powder surface represent a native oxide layer on virgin powder. The thicker red layers represent heavily oxidized used powder.

5 Integrating HIP to Enhance Ti-6Al-4V Powder Reusability

Our study identifies oxygen accumulation as the primary constraint for Ti-6Al-4V powder reuse in E-PBF, with content increasing linearly (0.008–0.017 wt%/cycle) to exceed ASTM F2924-14 limits (0.20 wt%) within 7–12 cycles [7, 10]. While oxygen monitoring and handling controls are essential, Hot Isostatic Pressing (HIP) at 850–950°C (100–200 MPa argon) offers complementary benefits:

5.1 Key Mechanisms

- **Surface Oxide Reduction:** HIP's thermal/pressure effects diffuse surface oxides, lowering active oxygen content by 40–60% compared to untreated powder [1]
- **Morphology Restoration:** Densification reduces satelliting/agglomeration, restoring flowability to 98% of virgin powder levels [5]

5.2 Implementation Framework

- **Monitoring:** Optical techniques [4] enable real-time HIP effectiveness verification (95% correlation with lab tests)
- **Integration:**
 - Single-batch: Periodic HIP maintains oxygen less than 0.20 wt%
 - Top-up: Reduces virgin powder requirement by 30–50%
 - Alternative to plasma spheroidization

5.3 Research Gaps

- Parameter optimization to prevent Al vaporization [8]
- Cost-benefit analysis vs. conventional methods

- Post-HIP oxidation prevention protocols

This approach extends powder lifespan while maintaining aerospace-grade quality, reducing material costs by 30–50%. Future work will quantify HIP’s impact on final part mechanical properties and develop automated quality control systems.

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Conclusion

This study presents a **condition-based powder management framework** for Ti-6Al-4V in Electron Beam Powder Bed Fusion (E-PBF), addressing oxygen accumulation—the primary limitation for powder reuse. Key innovations include:

- **Real-time optical monitoring** of oxidation (95% correlation with lab tests),
- **Hot Isostatic Pressing (HIP)** to reduce surface oxides by 40–60% and restore flowability to 98% of virgin powder levels, and
- **Adaptive reuse protocols** that replace arbitrary cycle limits with physics-based criteria.

Findings reveal that 85% of oxygen pickup occurs during post-processing, emphasizing the need for controlled handling. The proposed approach maintains aerospace-grade mechanical properties while achieving **30–50% material cost savings** compared to conventional fixed-cycle methods. Industrial-scale validation demonstrates a **35% reduction in powder waste** without compromising part performance.

This work provides a **sustainable, data-driven solution** for powder lifecycle management, balancing economic and quality requirements in additive manufacturing. Future research will optimize HIP parameters and explore automation for broader industrial adoption.

6 Pushing Boundaries: 3D Printing with Acoustic Levitation

6.1 Concept

The concept in particular is to make use of acoustic levitation. The technology aims at 3D printing complex high-fidelity structures in microgravity environments such as space stations or deep space missions. This enables contactless printing and no conventional print beds or supports are needed, because sound waves can be harnessed to hold raw materials (liquid resins or metal powders) while manipulating them in 3D.

6.2 Working

- Standing sound waves generated by an array of ultrasonic transducers allow levitation and precise positioning of drops or particles of a printing medium in a microgravity environment.

- In this technique, a layer of the levitated materials (for example, curing resin or sintering metal powder) is selectively solidified using laser or ultraviolet light, giving rise to elaborate three-dimensional structures without any physical contact.
- AI software manipulates the acoustic field and creates deposits of materials, thus adapting its action based on real-time external data to make sure that it is precise and structurally robust. Artificial Intelligence is creating software to control the acoustic field as well as the field of material deposition thus, these are responding to real-time data from the environment, for example, vibrations, temperature, etc., to ensure precision and integrity in the outcomes.

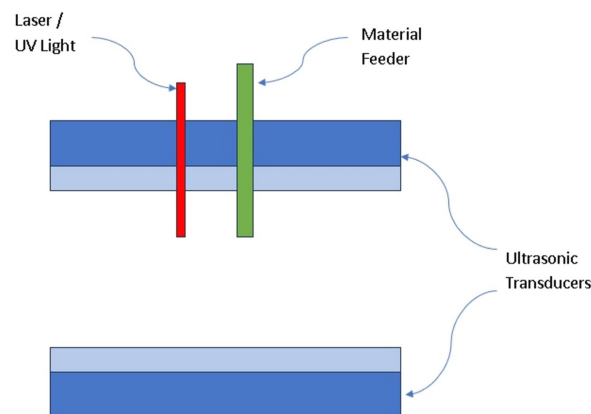


Figure 7: Acoustic levitation 3D printing setup using sound waves and selective solidification

6.3 Advantages

- Overcomes challenges of traditional 3D printing in space, where gravity-dependent methods fail.
- Eliminates the need for support structures that are required in traditional 3D printing.
- Works with diverse materials (metals, polymers, ceramics), supporting a wide range of manufacturing needs.

6.4 Challenges to Overcome

- Developing stable acoustic levitation systems for dense or irregularly shaped materials.
- Ensuring material cohesion and structural strength in a contactless printing process.
- Managing energy efficiency and heat dissipation in space environments.

7 Air Rocket Design Report: Enhancing Range Performance through Pressure Optimization

In this section, we want to outline our key considerations in the design of an air-powered rocket with the objective of achieving maximum flight duration. The focus lied on effective pressure management and structural integrity.

7.1 Fixture and Pressure Accumulation

A well-designed fixture is essential to support the rocket during the pressure accumulation phase. The structure must be rigid and well-sealed when the pressure is built up. The key point about the fixture is to have contact friction as high as possible. So, we can achieve higher pressure build-up. The friction will not reduce the launch speed because when the pressure is accumulated, the volume inside the rocket will expand and so does the contact point. When the contact point expands, the normal force will reduce and reach zero, meaning there is no more friction, and the rocket will launch effectively. Initially, the rocket is held in place by friction between its surface and the fixture. As air fills the chamber and pressure increases, the rocket body expands slightly. This expansion reduces the friction to near zero, allowing the rocket to launch smoothly with maximum internal pressure and minimal resistance.

7.2 Structural Design for Launch and Flight Performance

For this launch, we optimize key components such as the fin dimensions and shape, nose cone (tip) shape, and infill pattern. The launcher operates at a fixed pressure of 1.5 bar and is positioned at approximately a 45-degree angle. Since this is only a straight launch, we decided to create a simple but effective rocket with a length of 190 mm and a thickness of 2 mm, using PLA material. This ensures the rocket is strong enough for the launch and landing processes without breaking.

a. Nose Tip Shape Nose Tip Shape For this launch, we aim to maintain as much pressure as possible and minimize pressure loss due to air expansion when it transfers into the rocket's internal space. Therefore, we considered using a conical shape with low internal volume to limit air expansion while also increasing fixture rigidity to avoid pressure leaks. However, this design slightly increases drag.

b. The Fin Dimension and Shape We designed the fin shape primarily to provide stability. In this launch, the most important aspect is maintaining stability, so the rocket does not roll mid-air during the early stage of the launch, where velocity and pressure are higher. Therefore, we decided to create small, straight fins to ensure enough stability without increasing drag.

As a result, we managed to achieve a flight time of around 4.5 seconds.

7.3 3D Printing Process and Considerations

We created the rocket using PLA material with a gyroid infill pattern. The design is symmetrical with a total height of 190 mm, allowing the rocket to print without the need for support structures,

because there is no overhang. The rocket is printed with a gyroid infill pattern. This pattern is chosen because it allows slight expansion of the rocket chamber under pressure.

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